

Lab Taks-3

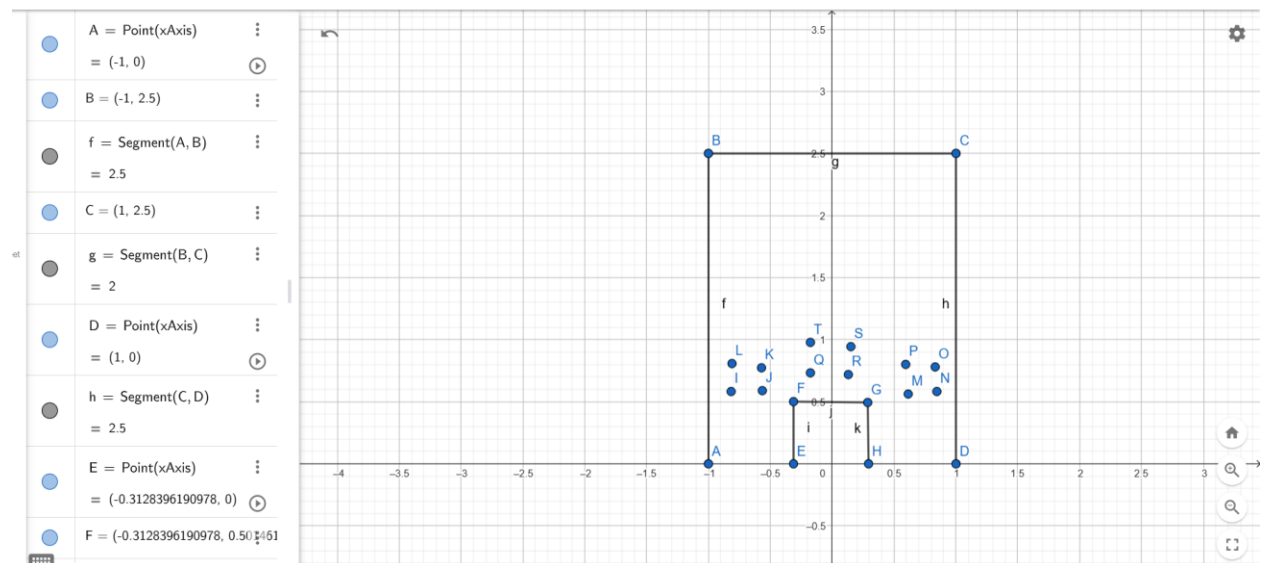
Submission Guidelines-

- Rename the file with your serial number only
- Must submit within time that will be discussed in class VUES
- Must include resources for all the section in the table

Question- 1

Draw five storied building with windows and a front door

Graph Plot (Picture)-



Code-

```
#include <windows.h>
#include <GL/glut.h>
```

```
void window(float x1, float y1, float x2, float y2)
{
    glBegin(GL_POLYGON);
    glColor3ub(173, 216, 230);
    glVertex2f(x1, y1);
    glVertex2f(x2, y1);
    glVertex2f(x2, y2);
    glVertex2f(x1, y2);
    glEnd();
}
```

```

    glLineWidth(2.0f);
    glBegin(GL_LINE_LOOP);
    glColor3ub(255, 255, 255);
    glVertex2f(x1, y1);
    glVertex2f(x2, y1);
    glVertex2f(x2, y2);
    glVertex2f(x1, y2);
    glEnd();
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);

    glBegin(GL_POLYGON);
    glColor3ub(46, 128, 202);
    glVertex2f(-1.5f, -1.0f);
    glVertex2f(1.5f, -1.0f);
    glVertex2f(1.5f, 7.7f);
    glVertex2f(-1.5f, 7.7f);
    glEnd();

    glBegin(GL_POLYGON);
    glColor3ub(139, 69, 19);
    glVertex2f(-0.5f, -1.0f);
    glVertex2f(0.5f, -1.0f);
    glVertex2f(0.5f, 2.0f);
    glVertex2f(-0.5f, 2.0f);
    glEnd();

    glLineWidth(3.0f);
    glBegin(GL_LINE_LOOP);
    glColor3ub(0, 0, 0);
    glVertex2f(-0.5f, -1.0f);
    glVertex2f(0.5f, -1.0f);
    glVertex2f(0.5f, 2.0f);
    glVertex2f(-0.5f, 2.0f);
    glEnd();

    for (float y = 2.5f; y <= 6.5f; y += 1.0f) {
        window(-1.3f, y, -0.5f, y + 0.6f);
    }
}

```

```
}

for (float y = 2.5f; y <= 6.5f; y += 1.0f) {
    window(0.5f, y, 1.3f, y + 0.6f);
}

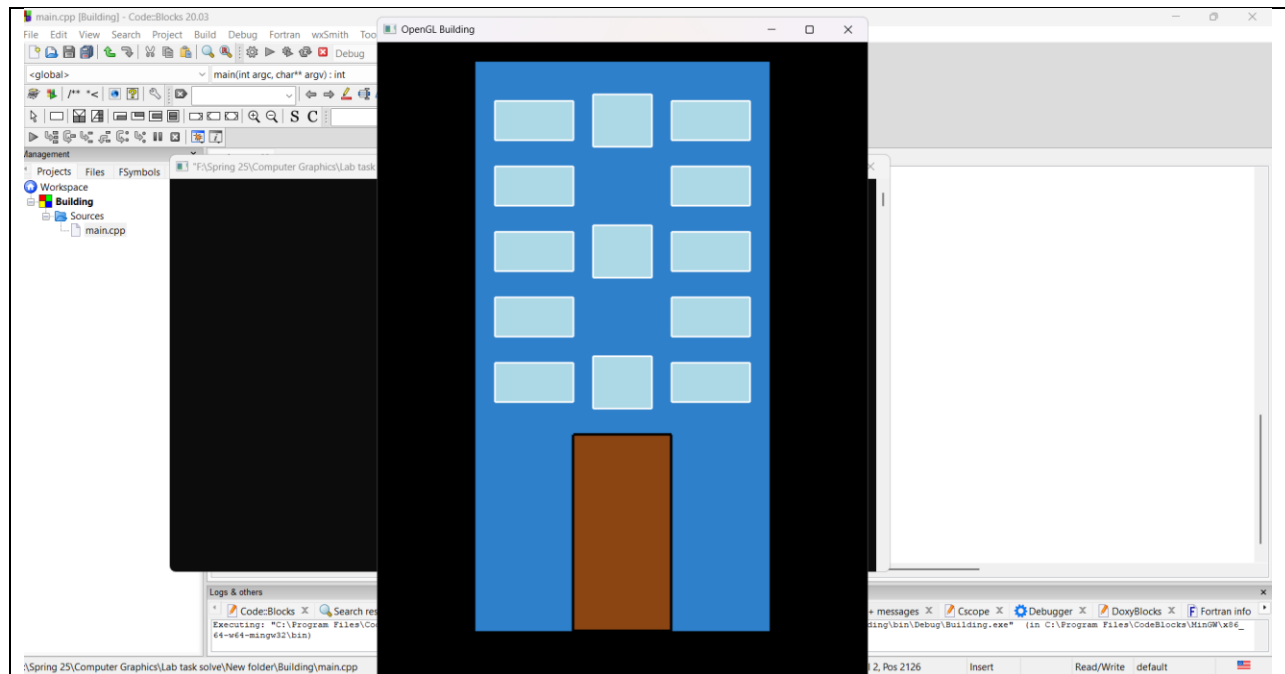
window(-0.3f, 6.4f, 0.3f, 7.2f);
window(-0.3f, 4.4f, 0.3f, 5.2f);
window(-0.3f, 2.4f, 0.3f, 3.2f);

glFlush();
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitWindowSize(600, 800);
    glutCreateWindow("OpenGL Building");

    gluOrtho2D(-2.5, 2.5, -2, 8); // Adjusted view
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}
```

Output Screenshot (Full Screen)-

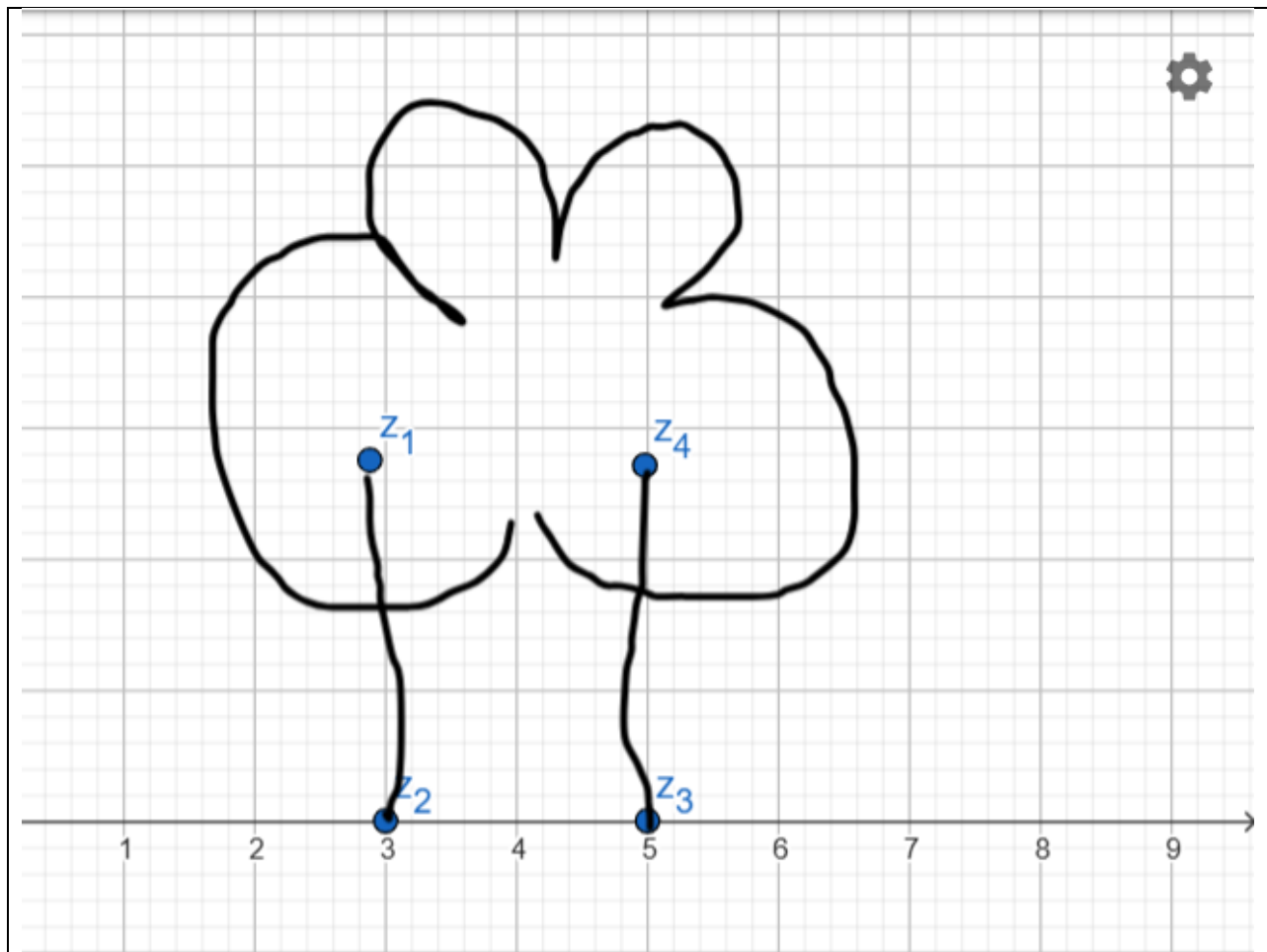


Question- 2

Draw a tree

Graph Plot (Picture)-

Resource – Youtube



Code-

```
#include <GL/glut.h>
```

```
#include <cmath>
```

```
void drawCircle(GLfloat cx, GLfloat cy, GLfloat r, int num_segments) {
    glBegin(GL_POLYGON);
    for (int i = 0; i < num_segments; i++) {
        float theta = 2.0f * M_PI * float(i) / float(num_segments);
        float x = r * cosf(theta);
        float y = r * sinf(theta);
        glVertex2f(cx + x, cy + y);
    }
    glEnd();
}
```

```
void init(void) {
    glClearColor(1.0, 1.0, 1.0, 1.0);
    glMatrixMode(GL_PROJECTION);
    gluOrtho2D(0, 400, 0, 500);
}
```

```

}

void display(void) {
    glClear(GL_COLOR_BUFFER_BIT);

    // Tree trunk
    glColor3ub(139, 69, 19);
    glBegin(GL_POLYGON);
    glVertex2f(180, 100);
    glVertex2f(220, 100);
    glVertex2f(220, 250);
    glVertex2f(180, 250);
    glEnd();

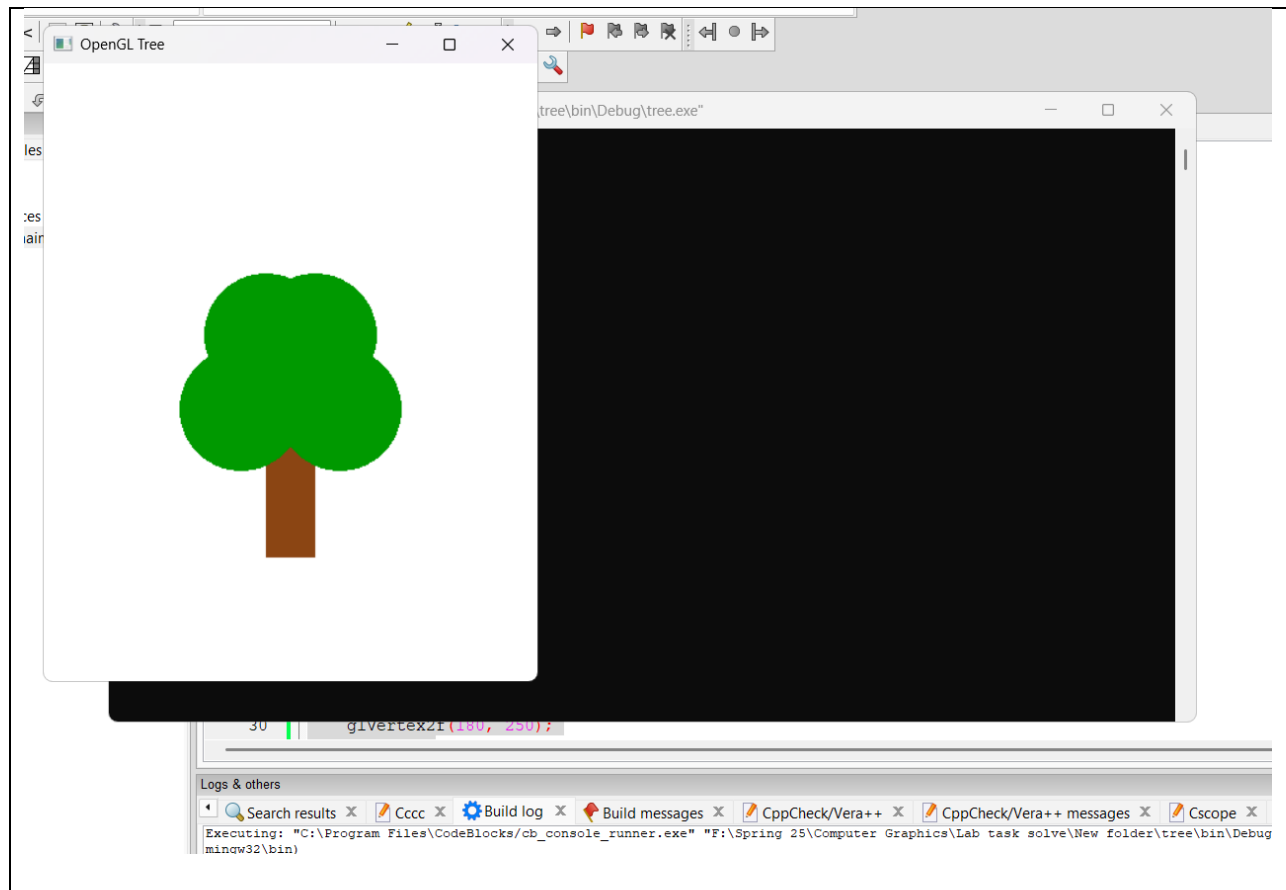
    // Tree leaves using circles
    glColor3ub(0, 153, 0);
    drawCircle(200, 250, 60, 50);
    drawCircle(160, 220, 50, 50);
    drawCircle(240, 220, 50, 50);
    drawCircle(180, 280, 50, 50);
    drawCircle(220, 280, 50, 50);

    glFlush();
    glutSwapBuffers();
}

int main(int argc, char **argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(400, 500);
    glutInitWindowPosition(100, 100);
    glutCreateWindow("OpenGL Tree");
    init();
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}

```

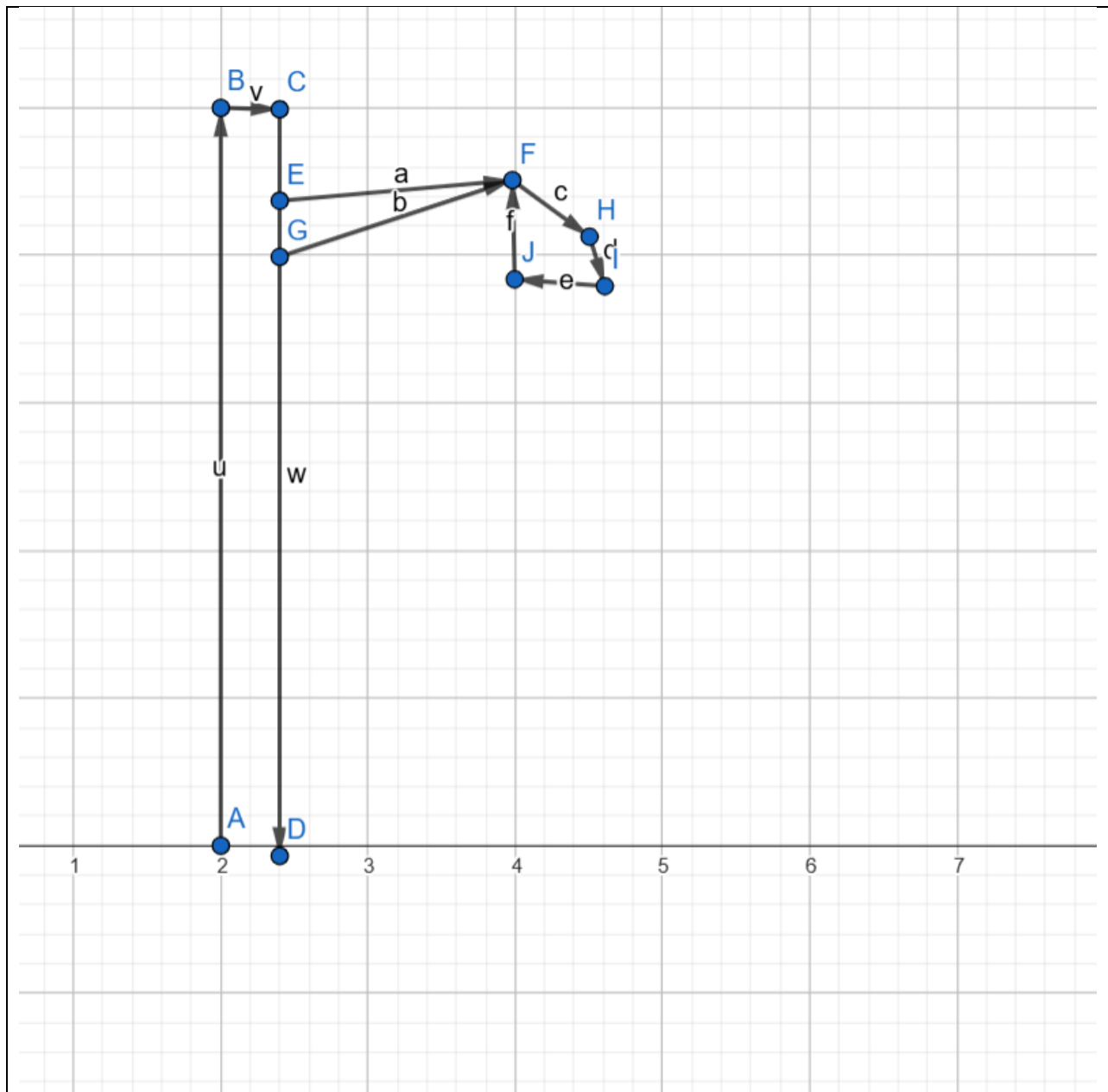
Output Screenshot (Full Screen)-



Question- 3

Draw a lamppost with black background

Graph Plot (Picture)-



Code-

```
#include <GL/glut.h>
```

```
void drawLamppost() {
    glClear(GL_COLOR_BUFFER_BIT);

    glColor3f(0.6, 0.6, 0.6);
    glBegin(GL_QUADS);
    glVertex2f(2.4, 0.0);
```



```

    glVertex2f(2.6, 0.0);
    glVertex2f(2.6, 4.5);
    glVertex2f(2.4, 4.5);
    glEnd();

    glColor3f(0.4, 0.4, 0.4);
    glBegin(GL_QUADS);
    glVertex2f(2.0, -0.2);
    glVertex2f(3.0, -0.2);
    glVertex2f(3.0, 0.0);
    glVertex2f(2.0, 0.0);
    glEnd();

    glColor3f(0.6, 0.6, 0.6);
    glBegin(GL_QUADS);
    glVertex2f(2.0, 4.5);
    glVertex2f(3.8, 4.5);
    glVertex2f(3.8, 4.3);
    glVertex2f(2.0, 4.3);
    glEnd();

    glColor3f(1.0, 1.0, 0.0);
    glBegin(GL_POLYGON);
    glVertex2f(3.5, 4.3);
    glVertex2f(4.0, 4.0);
    glVertex2f(4.2, 3.7);
    glVertex2f(3.8, 3.5);
    glVertex2f(3.3, 3.7);
    glEnd();

    glFlush();
}

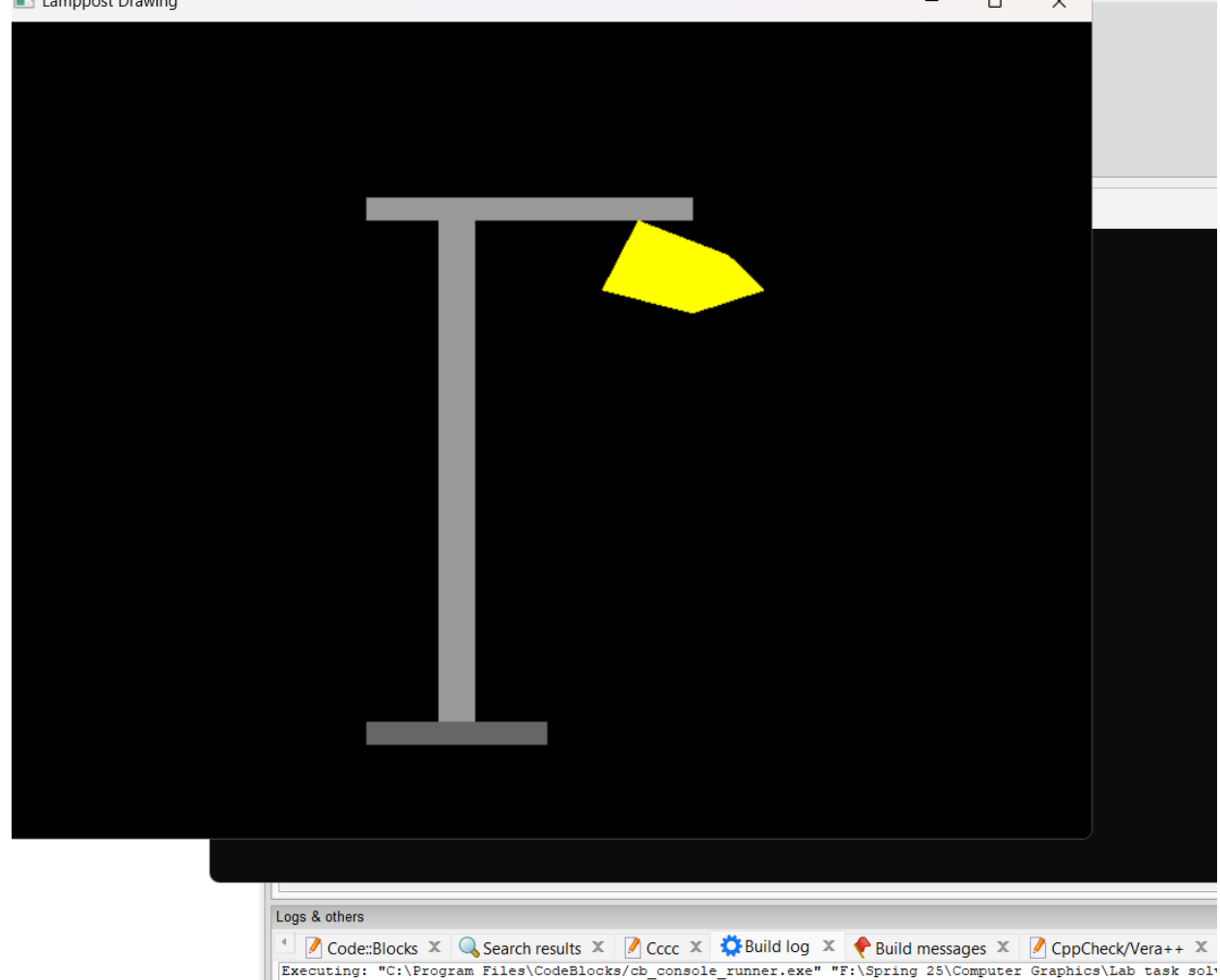
void init() {
    glClearColor(0.0, 0.0, 0.0, 1.0); // Black background
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(0.0, 6.0, -1.0, 6.0); // Adjusted coordinate system
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(800, 600);

```

```
glutCreateWindow("Lamppost Drawing");  
init();  
glutDisplayFunc(drawLamppost);  
glutMainLoop();  
return 0;  
}
```

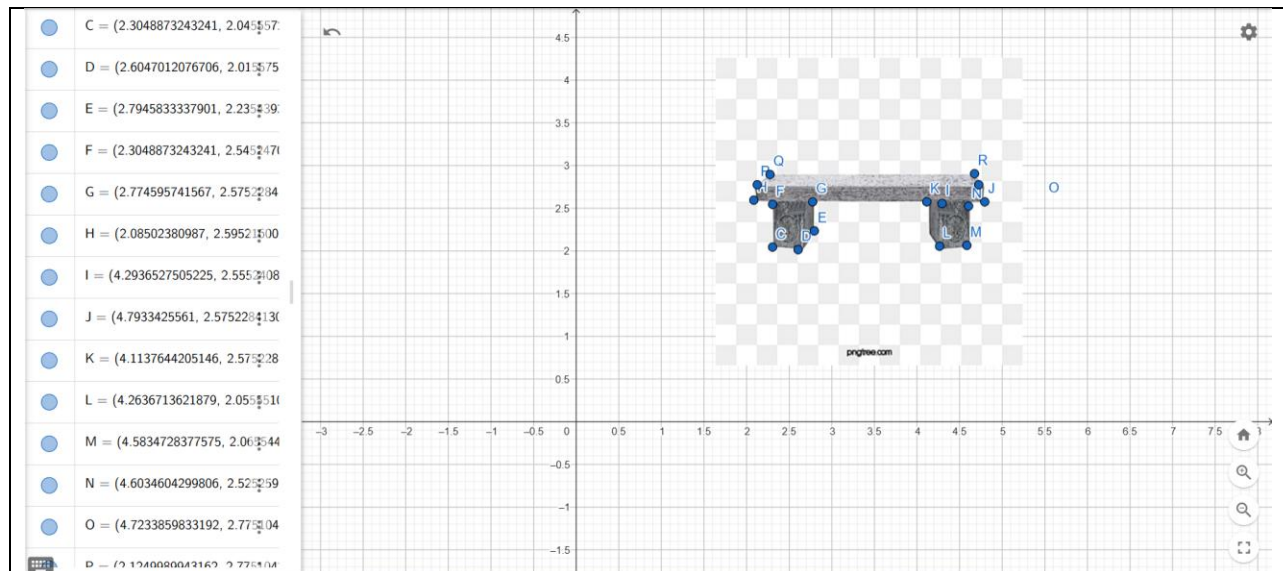
Output Screenshot (Full Screen)-



Question- 4

Draw a bench

Graph Plot (Picture)-



Code-

```
#include <windows.h>
```

```
#include <GL/glut.h>
```

```
// Adjusted Bench Coordinates (More structured)
```

```
float seat[] = {2.0, 3.0, 5.0, 3.0, 5.0, 2.7, 2.0, 2.7}; // Seat
```

```
float leftLeg[] = {2.3, 2.7, 2.5, 2.7, 2.5, 1.5, 2.3, 1.5}; // Left Leg
```

```
float rightLeg[] = {4.5, 2.7, 4.7, 2.7, 4.7, 1.5, 4.5, 1.5}; // Right Leg
```

```
float backrest[] = {2.0, 3.5, 5.0, 3.5, 5.0, 3.0, 2.0, 3.0}; // Backrest
```

```
void drawQuad(float coords[]) {
```

```
    glBegin(GL_QUADS);
```

```
    glVertex2f(coords[0], coords[1]);
```

```
    glVertex2f(coords[2], coords[3]);
```

```
    glVertex2f(coords[4], coords[5]);
```

```
    glVertex2f(coords[6], coords[7]);
```

```
    glEnd();
```

```
}
```

```
void drawBench() {
```

```
    // Draw the Bench Components
```

```
    glColor3f(0.6f, 0.6f, 0.6f); // Grey Seat
```

```
    drawQuad(seat);
```

```
    glColor3f(0.4f, 0.4f, 0.4f); // Darker Legs
```

```
    drawQuad(leftLeg);
```

```
    drawQuad(rightLeg);
```

```
    glColor3f(0.5f, 0.5f, 0.5f);
```

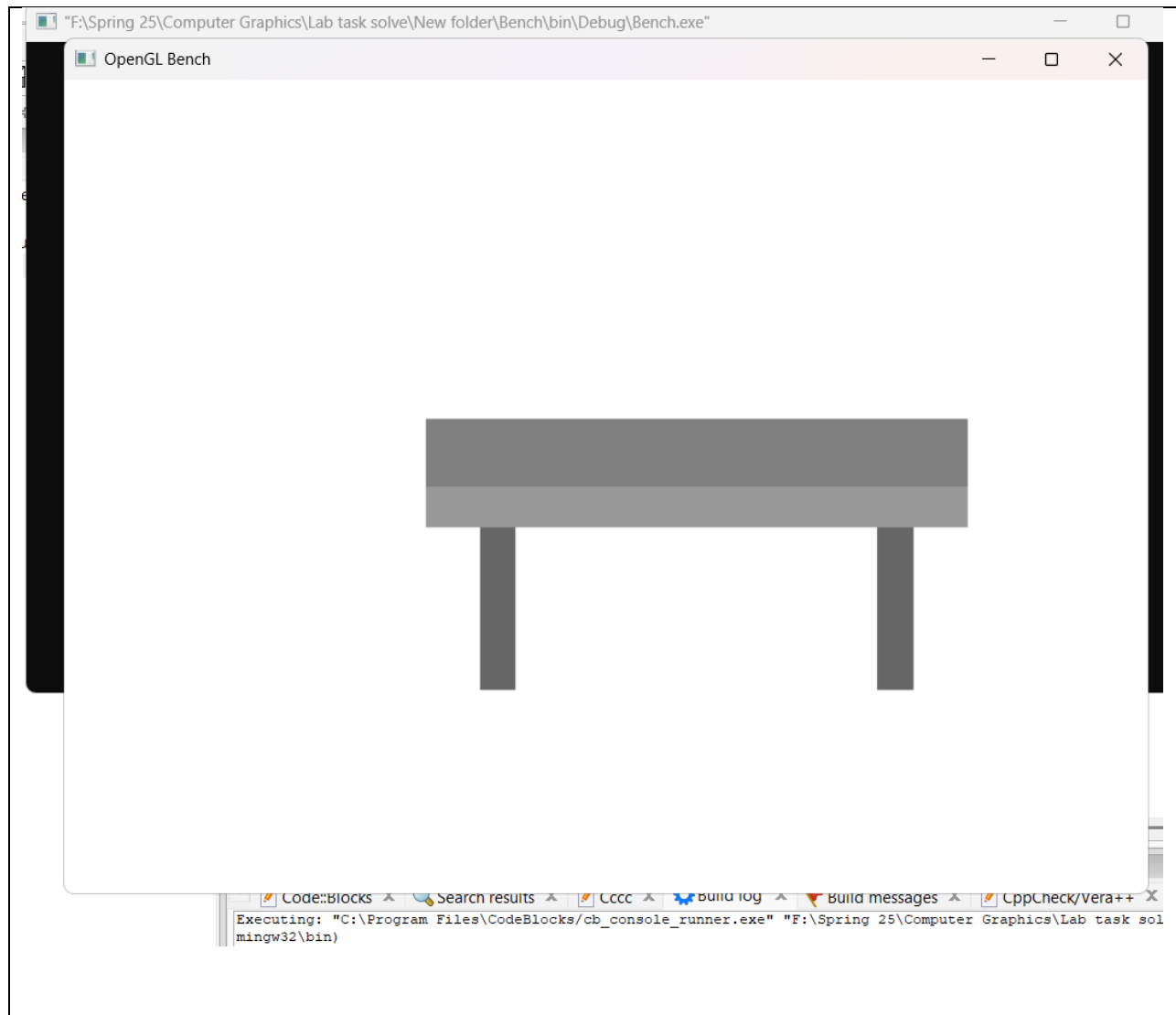
```
    drawQuad(backrest);
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);
    drawBench();
    glFlush();
}

void reshape(int w, int h) {
    glViewport(0, 0, w, h);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(0.0, 6.0, 0.0, 6.0);
    glMatrixMode(GL_MODELVIEW);
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(800, 600);
    glutCreateWindow("OpenGL Bench");
    glutDisplayFunc(display);
    glutReshapeFunc(reshape);
    glClearColor(1.0, 1.0, 1.0, 1.0);
    glutMainLoop();
    return 0;
}
```

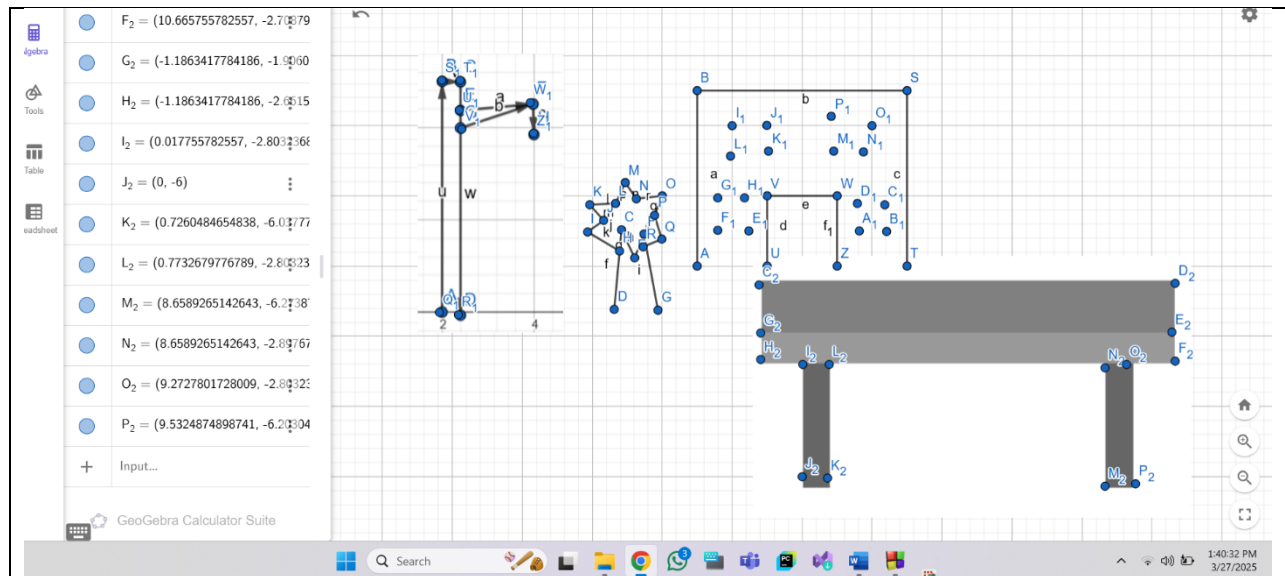
Output Screenshot (Full Screen)-



Question- 5

Use the building, tree, lamppost and bench to create a scenario

Graph Plot (Picture)-



```
Code-#include <windows.h>
#include <GL/glut.h>
#include <cmath>

// Draw a night-time background (dark sky and ground)
void drawBackground() {
    glClearColor(0.1, 0.1, 0.2, 1.0); // Dark blue sky for night
    glClear(GL_COLOR_BUFFER_BIT);

    glColor3ub(34, 34, 34); // Dark ground for night
    glBegin(GL_POLYGON);
    glVertex2f(-4.0, -1.0);
    glVertex2f(4.0, -1.0);
    glVertex2f(4.0, 1.0);
    glVertex2f(-4.0, 1.0);
    glEnd();
}

// Draw a circle for tree leaves
void drawCircle(GLfloat cx, GLfloat cy, GLfloat r, int num_segments) {
    glBegin(GL_POLYGON);
    for (int i = 0; i < num_segments; i++) {
        float theta = 2.0f * M_PI * float(i) / float(num_segments);
        float x = r * cosf(theta);
        float y = r * sinf(theta);
        glVertex2f(cx + x, cy + y);
    }
    glEnd();
}
```

```

// Draw the 5-story building with windows
void drawBuilding() {
    glColor3ub(46, 128, 202); // Light building color
    glBegin(GL_POLYGON);
    glVertex2f(-1.5f, -1.0f);
    glVertex2f(1.5f, -1.0f);
    glVertex2f(1.5f, 7.0f); // 5 stories building
    glVertex2f(-1.5f, 7.0f);
    glEnd();

    // Windows (5 floors of windows with dim light)
    glColor3ub(173, 216, 230); // Soft light blue for night windows
    for (float y = 1.0f; y <= 5.5f; y += 1.2f) {
        for (float x = -1.2f; x <= 1.0f; x += 1.2f) {
            glBegin(GL_POLYGON);
            glVertex2f(x, y);
            glVertex2f(x + 0.5f, y);
            glVertex2f(x + 0.5f, y + 0.6f);
            glVertex2f(x, y + 0.6f);
            glEnd();
        }
    }
}

// Draw the tree with circular leaves behind the building
void drawTree() {
    // Tree trunk (placed behind the building)
    glColor3ub(139, 69, 19); // Darker trunk color for night
    glBegin(GL_POLYGON);
    glVertex2f(-2.0, -1.0);
    glVertex2f(-1.7, -1.0);
    glVertex2f(-1.7, 3.5); // Taller trunk
    glVertex2f(-2.0, 3.5); // Taller trunk
    glEnd();

    // Tree leaves (circle-like foliage)
    glColor3ub(0, 153, 0); // Dark green for leaves at night
    drawCircle(-2.0, 4.5, 1.5, 50);
    drawCircle(-1.5, 5.5, 1.2, 50);
    drawCircle(-2.5, 5.5, 1.2, 50);
    drawCircle(-2.0, 6.5, 1.0, 50);
    drawCircle(-1.5, 6.5, 1.0, 50);
}

```

```
// Draw the lamppost beside the bench with night-time lighting
```

```
void drawLamppost() {
```

```
    glColor3f(0.6, 0.6, 0.6); // Lamppost color
```

```
    glBegin(GL_QUADS);
```

```
    glVertex2f(1.7, 0.0);
```

```
    glVertex2f(1.8, 0.0);
```

```
    glVertex2f(1.8, 4.5);
```

```
    glVertex2f(1.7, 4.5);
```

```
    glEnd();
```

```
    glColor3f(0.6, 0.6, 0.6); // Top of lamppost
```

```
    glBegin(GL_QUADS);
```

```
    glVertex2f(1.5, 4.5);
```

```
    glVertex2f(2.0, 4.5);
```

```
    glVertex2f(2.0, 4.3);
```

```
    glVertex2f(1.5, 4.3);
```

```
    glEnd();
```

```
// Light (circular) with bright yellow for night-time glow
```

```
glColor3f(1.0, 1.0, 0.0); // Yellow light
```

```
float radius = 0.3;
```

```
int num_segments = 30;
```

```
glBegin(GL_POLYGON);
```

```
for (int i = 0; i < num_segments; i++) {
```

```
    float theta = 2.0f * M_PI * float(i) / float(num_segments);
```

```
    float x = radius * cosf(theta);
```

```
    float y = radius * sinf(theta);
```

```
    glVertex2f(1.85f + x, 4.3f + y);
```

```
}
```

```
glEnd();
```

```
}
```

```
// Draw the bench on the right side of the scene
```

```
void drawBench() {
```

```
    glColor3f(0.6f, 0.6f, 0.6f); // Bench seat color
```

```
    glBegin(GL_QUADS);
```

```
    glVertex2f(0.5, -0.8); // Moved to the right
```

```
    glVertex2f(3.5, -0.8); // Moved to the right
```

```
    glVertex2f(3.5, -1.0); // Moved to the right
```

```
    glVertex2f(0.5, -1.0); // Moved to the right
```

```
    glEnd();
```

```
    glColor3f(0.4f, 0.4f, 0.4f); // Bench legs color
```



```

glBegin(GL_QUADS);
glVertex2f(0.6, -1.0); // Left leg
glVertex2f(0.7, -1.0); // Left leg
glVertex2f(0.7, -1.3); // Left leg
glVertex2f(0.6, -1.3); // Left leg
glEnd();

glBegin(GL_QUADS);
glVertex2f(3.4, -1.0); // Right leg
glVertex2f(3.5, -1.0); // Right leg
glVertex2f(3.5, -1.3); // Right leg
glVertex2f(3.4, -1.3); // Right leg
glEnd();
}

void display() {
    drawBackground(); // Dark sky and ground for night-time
    drawBuilding();   // Building with dimly lit windows
    drawTree();       // Tree with circular leaves behind the building
    drawLamppost();   // Lamppost with lighting
    drawBench();      // Bench moved to the right side
    glFlush();
}

void init() {
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(-4.0, 4.0, -2.0, 8.0);
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(800, 600);
    glutCreateWindow("OpenGL Night Scene");
    init();
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}

```

Output Screenshot (Full Screen)-

